



FPDAF (Federated Personalized Drift Alerting Framework): CLINICAL AI EARLY-WARNING WORKSTATION

Explainable Federated Learning for Sepsis CDSS with Personalized Drift Adaptation

MSME Idea Hackathon 6.0 Proposal | **Amrita Vishwa Vidyapeetham**

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The Clinical Sepsis Crisis & Privacy Bottleneck

[Crisis] National Mortality Burden in India

- ▶ **2.9 to 3.0 Million Deaths** annually in India (accounting for **30%** of total national deaths).
- ▶ **34% to 50%+ Mortality** in Indian ICUs.
- ▶ **8% Survival Drop** per hour of delayed detection.

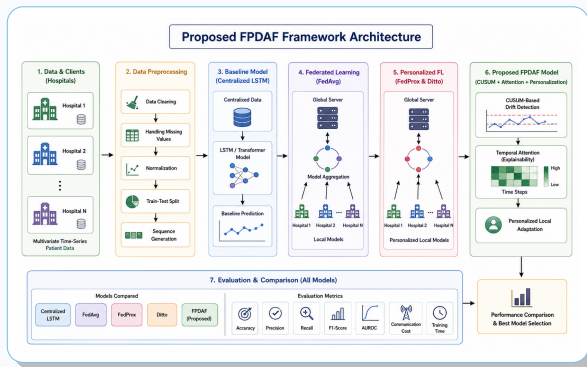
[Bottleneck] Data Sovereignty & Drift Limits

- ▶ **DPDPA & HIPAA Privacy Mandates:** Private patient records cannot be centralized to external cloud servers.
- ▶ **Clinical Data Drift:** Hospital demographics vary, causing static cloud AI models to lose accuracy.

The FPDFAF Solution

FPDFAF acts as a bedside early-warning radar system using Federated Learning to train collaborative models while keeping raw patient data localized inside hospital firewalls.

Proposed FPDAF Solution & System Architecture



System Dataflow: Local ICU vital streams pass through a shared BiLSTM backbone and DDP cross-attention layers. Local CUSUM triggers activate Client-Side Selective Personalization (CSSP) upon demographic drift without uploading raw EHR records.

Key Technical Novelties

- ▶ **1. Dynamic Dual-Phase (DDP) Attention:** 2D cross-attention ($\alpha_t \otimes \beta_v$) tracking 24h sequence hours AND vital co-dependencies.
- ▶ **2. AD-CUSUM Drift Auditing:** Online loss residual control charts flagging hospital demographic shifts in real-time.
- ▶ **3. Client-Side Selective Personalization (CSSP):** Freezes shared LSTM backbones, saving **38% bandwidth** & adapting in **6 mins**.

Competitive Comparison

Feature	FPDAF	Epic	Philips
Federated Privacy	Yes	No	No
Concept Drift Auditing	Yes	No	No
Temporal XAI Heatmaps	Yes	No	No
Edge-AI Deployment	Yes	Partial	Yes

Experimental Performance Metrics

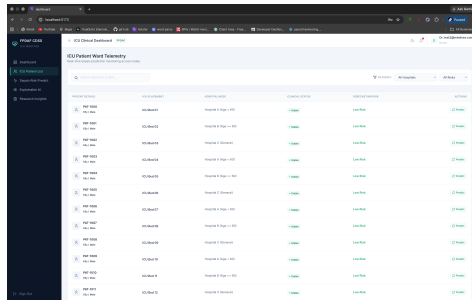
Validated on multi-center PhysioNet ICU cohorts:

- ▶ **86.51% Accuracy:** Superior to FedAvg (75.94%) and FedProx (78.85%).
- ▶ **0.8214 AUROC:** High predictive discrimination.
- ▶ **38% Bandwidth Savings:** 1.52 GB vs 2.48 GB.

Software Stack Operational

FastAPI backend, SQLite relational warehouse ('clinical_warehouse.db'), and React Vite bedside dashboard fully functional.

Clinician Bedside UI Workspace



The screenshot shows a web application titled "ICU Patient Ward Teleretry". It features a sidebar with navigation options: "PROFIT CODE", "Dashboard", "ICU Patient Ward Teleretry", "ICU Patient Ward Teleretry", "ICU Patient Ward Teleretry", and "ICU Patient Ward Teleretry". The main content area displays a table with columns: "PATIENT ID", "ICU WARD", "WARD RISK", "CLINICAL STATUS", "WARD RISK", and "ACTION". The table lists 10 patients, each with a unique ID, ward, risk level, and clinical status. The "ACTION" column contains a link icon.

PATIENT ID	ICU WARD	WARD RISK	CLINICAL STATUS	WARD RISK	ACTION
PAT 0001	ICU Ward 1	High Risk	Stable	Low Risk	[Link]
PAT 0002	ICU Ward 2	High Risk	Stable	Low Risk	[Link]
PAT 0003	ICU Ward 3	High Risk	Stable	Low Risk	[Link]
PAT 0004	ICU Ward 4	High Risk	Stable	Low Risk	[Link]
PAT 0005	ICU Ward 5	High Risk	Stable	Low Risk	[Link]
PAT 0006	ICU Ward 6	High Risk	Stable	Low Risk	[Link]
PAT 0007	ICU Ward 7	High Risk	Stable	Low Risk	[Link]
PAT 0008	ICU Ward 8	High Risk	Stable	Low Risk	[Link]
PAT 0009	ICU Ward 9	High Risk	Stable	Low Risk	[Link]
PAT 0010	ICU Ward 10	High Risk	Stable	Low Risk	[Link]

Doctor Portal: Hourly vitals, risk dials, and temporal XAI heatmaps.

Detailed Budget Breakup & Justification (Total: Rs. 15.0 Lakhs)

Financial Outlay Under MSME Idea Hackathon 6.0 Guidelines (Maximum Grant Cap):

Comprehensive justification for Rs. 15 Lakhs to build physical Edge-AI bedside nodes and execute clinical trials:

1. Technology Costs (10.0L)

- ▶ **Hardware & Assembly (3.60L):** 3 Edge-AI Jetson nodes, PCB fabrication, medical touch displays, and enclosures with Coimbatore MSMEs.
- ▶ **Data Curation (3.40L):** Expert clinician annotation & vital stream simulation.
- ▶ **Security Audits (3.00L):** Penetration testing & DPDPA compliance audits.

2. Mentorship (3.0L)

- ▶ **Clinical Guidance (1.80L):** ICU specialist honorarium for clinical protocol validation.
- ▶ **Lab Support (1.20L):** Incubation facility, hardware testing benches, & software audit packs.

3. Field Trials (2.0L)

- ▶ **Node Deployment (1.00L):** On-site installation across partnering regional ICUs.
- ▶ **Telemetry Audits (1.00L):** Multi-site ICU field data collection and travel.

Indian MSME Hardware Integration

FPDAF is an **integrated medical hardware product**:

- ▶ **Physical Node Enclosure:** Bedside cabinet with custom display, visual alarms, and network ports.
- ▶ **Local MSME Manufacturing:** Partnering with Coimbatore electronics MSMEs for PCB assembly and cabinet fabrication.

12-Month Commercialization Roadmap

1. **Months 1-2:** Edge PCB fabrication & MSME enclosure assembly.
2. **Months 3-5:** Multi-node network setup & shadow trials.
3. **Months 6-12:** ISO 13485 / DPDPA audit pack & CDSCO Class B filing.

Thank You

FPDAF: Explainable Edge-AI to Save Lives at the Bedside

MSME Idea Hackathon 6.0

Sub-Theme 6: Frontier Technologies | **Amrita Vishwa Vidyapeetham**

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Questions & Expert Discussion